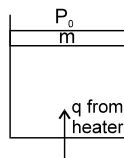


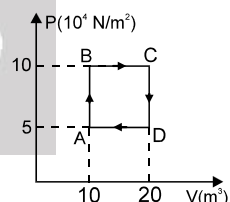


High Level Problems (HLP)

SUBJECTIVE QUESTIONS

- A vessel of volume V is evacuated by means of a piston air pump. In one stroke the piston is pulled out to make the volume of gas $V + \Delta V$ then ΔV volume from this is taken out leaving volume V in the cylinder. How many strokes are needed to reduce the pressure in the vessel to $1/\eta$ times the initial pressure? The process is assumed to be isothermal, and the gas is an ideal.
- Find the pressure of air in a vessel being evacuated as a function of evacuation time t . The vessel volume is V , the initial pressure is p_0 . The process is assumed to be isothermal, and the evacuation rate equal to C and independent of pressure.
Note: The evacuation rate is the gas volume being evacuated per unit time, with that volume being measured under the gas pressure attained by that moment.
- Find the maximum attainable temperature of an ideal gas in the following process : where (a) $p = p_0 - \alpha V^2$; (b) $p = p_0 e^{-\beta V}$, where p_0 , α and β are positive constants and V is the volume of one mole of gas.
- Two moles of an ideal monoatomic gas are contained in a vertical cylinder of cross sectional area A as shown in the figure. The piston is frictionless and has a mass m . At a certain instant a heater starts supplying heat to the gas at a constant rate q J/s. Find the steady velocity of the piston under isobaric condition. All the boundaries are thermally insulated.


- A piston can freely move inside a horizontal cylinder closed from both ends. Initially, the piston separates the inside space of the cylinder into two equal parts each of volume V_0 , in which an ideal gas is contained under the same pressure p_0 and at the same temperature. What work has to be performed in order to increase isothermally the volume of one part of gas η times compared to that of the other by slowly moving the piston?
- At 27°C two moles of an ideal monoatomic gas occupy a volume V . The gas expands adiabatically to a volume $2V$. Calculate :
 (i) the final temperature of the gas, (ii) change in its internal energy and (iii) the work done by the gas during the process. (Take $R = \frac{25}{3}$ J/mol-K)
[JEE 1996, 5/100]
- A vertical hollow cylinder contains an ideal gas. The gas is enclosed by a 5kg movable piston with an area of cross-section $5 \times 10^{-3} \text{ m}^2$. Now, the gas is heated slowly from 300 K to 350 K and the piston rises by 0.1 m. The piston is now clamped at this position and the gas is cooled back to 300 K. Find the difference between the heat energy added during heating process and energy lost during the cooling process. [1 atm pressure = 10^5 N m^{-2}]
[JEE 1996, 5]
- A sample of 2 kg of monoatomic Helium (assumed ideal) is taken through the process ABC and another sample of 2 kg of the same gas is taken through the process ADC as in figure. Given, molecular mass of Helium = 4.
 (i) What is the temperature of Helium in each of the states A, B, C and D ?
 (ii) Is there any way of telling afterwards which sample of Helium went through
 (iii) How much is the heat involved in each of the processes ABC and ADC.
[JEE 1997, 5/100]


- Two moles of an ideal monoatomic gas are confined within a cylinder by a massless and frictionless spring loaded piston of cross-sectional area $4 \times 10^{-3} \text{ m}^2$. The spring is, initially in its relaxed state. Now the gas is heated by an electric heater, placed inside the cylinder, for some time. During this time, the gas expands and does 50 J of work in moving the piston through a distance 0.10 m. The temperature of the gas increases by 50 K. Calculate the spring constant and the heat supplied by the heater.
 $P_{\text{atm}} = 1 \times 10^5 \text{ N/m}^2$. $R = 8.314 \text{ J/mol-K}$
[JEE 1997, 5]
- Two vessels A and B, thermally insulated, contain an ideal monoatomic gas. A small tube fitted with a valve connects these vessels. Initially the vessel A has 2 liters of gas at 300 K and $2 \times 10^5 \text{ N m}^{-2}$ pressure while vessel B has 4 liters of gas at 350 K and $4 \times 10^5 \text{ N m}^{-2}$ pressure. The valve is now opened and the system reaches equilibrium in pressure and temperature. Calculate the new pressure and temperature. ($R = \frac{25}{3}$ J/mol-K)
[JEE 1997, 5]

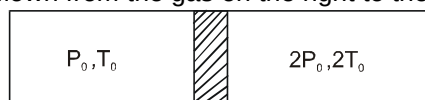
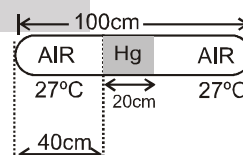
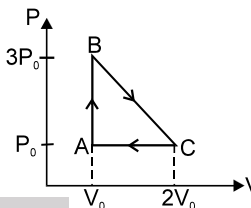
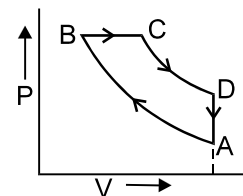


11. One mole of a diatomic ideal gas ($\gamma = 1.4$) is taken through a cyclic process starting from point A. The process $A \rightarrow B$ is an adiabatic compression. $B \rightarrow C$ is isobaric expansion. $C \rightarrow D$ an adiabatic expansion and $D \rightarrow A$ is isochoric as shown in P-V diagram. The volume ratios are $\frac{V_A}{V_B} = 16$ & $\frac{V_C}{V_B} = 2$ and the temperature at A is $T_A = 300$ K. Calculate the temperature of the gas at the points B and D and find the efficiency of the cycle. [JEE 1997, 5/100]
12. One mole of an ideal monoatomic gas is taken around the cyclic process ABCA as shown in the figure. Calculate, [JEE 1998, 8/200]
 (a) the work done by the gas ;
 (b) the heat rejected by the gas in the path CA and the heat absorbed by the gas in the path AB ;
 (c) the net heat absorbed by the gas in the path BC ;
 (d) the maximum temperature attained by the gas during the cycle.
13. Two moles of a monatomic gas, initially at pressure P_1 and volume V_1 , undergo an adiabatic compression until its volume becomes V_2 . Then the gas is given heat Q at constant volume V_2 .
 (a) Sketch the complete process on a P-V diagram.
 (b) Find the total work done by the gas, the total change in its internal energy and the final temperature of the gas. [Give your answers in terms of P_1 , V_1 , V_2 and Q and R .] [JEE 1999, 2 + 8 / 200]
14. A weightless piston divides a thermally insulated cylinder into two parts of volumes V and $3V$. 2 moles of an ideal gas at pressure $P = 2$ atmosphere are confined to the part with volume $V = 1$ liter. The remainder of the cylinder is evacuated. The piston is now released and the gas expands to fill the entire space of the cylinder. The piston is then pressed back to the initial position. Find the increase of internal energy in the process and final temperature of the gas. The ratio of the specific heats of the gas, $\gamma = 1.5$. [REE -1999]
15. Two containers A and B of equal volume $V_0/2$ each are connected by a narrow tube which can be closed by a valve. The containers are fitted with pistons which can be moved to change the volumes. Initially, the valve is open and the containers contain an ideal gas ($C_p/C_v = \gamma$) at atmospheric pressure P_0 and atmospheric temperature $2T_0$. The walls of the containers A are highly conducting and of B are non-conducting. The valve is now closed and the pistons are slowly pulled out to increase the volumes of the containers to double the original value. (a) Calculate the temperatures and pressures in the two containers. (b) The valve is now opened for sufficient time so that the gases acquire a common temperature and pressure. Find the new values of the temperature and the pressure.
16. In given figure, an adiabatic cylindrical tube of volume $2V_0$ is divided in two equal parts by a frictionless adiabatic separator. An ideal gas in left side of a tube having pressure P_1 and temperature T_1 where as in the right side having pressure P_2 and temperature T_2 . $C_p/C_v = \gamma$ is the same for both the gases. The separator is slid slowly and is released at a position where it can stay in equilibrium. Find (a) the final volumes of the two parts, (b) the heat given to the gas in the left part and (c) the final common pressure of the gases.
- P_1, T_1

P_2, T_2
17. In the given figure a glass tube lies horizontally with the middle 20 cm containing mercury. The two ends of the tube contains air at 27°C and at a pressure 76 cm of mercury. Now the air column on one side is maintained at 0°C and the other side is maintained at 127°C . Find the new length of the air column on the glass. Neglect the changes in the volume of mercury and of the glass.
- P_0, T_0

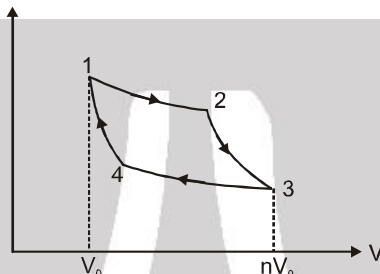
$2P_0, 2T_0$

adiabatic
18. A cylindrical tube with adiabatic walls having volume $2V_0$ contains an ideal monoatomic gas as shown in figure. The tube is divided into two equal parts by a fixed super conducting wall. Initially, the pressure and the temperature are P_0, T_0 on the left and $2P_0, 2T_0$ on the right. When system is left for sufficient amount of time the temperature on both sides becomes equal (a) Find work done by the gas on the right part? (b) Find the final pressures on the two sides. (c) Find the final equilibrium temperature. (d) How much heat has flown from the gas on the right to the gas on the left?





19. An ideal gas ($C_p/C_v = \gamma$) having initial pressure P_0 and volume V_0 . (a) The gas is taken isothermally to a pressure $2P_0$ and then adiabatically to a pressure $4P_0$. Find the final volume. (b) The gas is brought back to its initial state. It is adiabatically taken to a pressure $2P_0$ and then isothermally to a pressure $4P_0$. Find the final volume.
20. Two samples A and B of the same gas have equal volumes and pressures. The gas in sample A is expanded isothermally to four times of its initial volume and the gas in B is expanded adiabatically to double its volume. If work done in isothermal process is twice that of adiabatic process, then show that γ satisfies the equation $1 - 2^{1-\gamma} = (\gamma-1) \ln 2$.
21. A Carnot engine cycle is shown in the Fig. (2). The cycle runs between temperatures $T_H = \alpha T_0$ and $T_L = T_0$ ($\alpha > 1$). Minimum and maximum volume at state 1 and state 3 are V_0 and nV_0 respectively. The cycle uses one mole of an ideal gas with $C_p/C_v = \gamma$. Here C_p and C_v are the specific heats at constant pressure and volume respectively. You must express all answers in terms of the given parameters $\{\alpha, n, T_0, V_0, ?\}$ and universal gas constant R . [Olympiad 2011]



- (a) Find P, V, T for all the states
 (b) Calculate the work done by the engine in each process: $W_{12}, W_{23}, W_{34}, W_{41}$.
 (c) Calculate Q , the heat absorbed in the cycle.
22. A thermally insulated piston divides a nonconducting container in two compartments, right compartment of $2V$, T and $2P$, while in the left compartment the respective values are V , T and P . Total moles in total system of both compartments is 5 moles (same molar mass). If the piston can slide freely, and in the final equilibrium position, volume of right compartment is $\frac{xV}{5}$ then find the value of x .
23. **Cloud formation condition**
 Consider a simplified model of cloud formation. Hot air in contact with the earth's surface contains water vapor. This air rises convectively till the water vapor content reaches its saturation pressure. When this happens, the water vapor starts condensing and droplets are formed. We shall estimate the height at which this happens. We assume that the atmosphere consists of the diatomic gases oxygen and nitrogen in the mass proportion 21:79 respectively. We further assume that the atmosphere is an ideal gas, g the acceleration due to gravity is constant and air processes are adiabatic. Under these assumptions one can show that the pressure is given by [OLYMPIAD 2012]

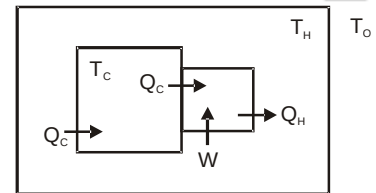
$$p = p_0 \left(\frac{T_0 - \Gamma z}{T_0} \right)^\alpha$$

Here p_0 and T_0 is the pressure and temperature respectively at sea level ($z = 0$), Γ is the lapse rate (magnitude of the change in temperature T with height z above the earth's surface, i.e. $\Gamma > 0$).

- (a) Obtain an expression for the lapse rate Γ in terms of γ, R, g and m_a . Here γ is the ratio of specific heat at constant pressure to specific heat at constant volume; R , the gas constant; and m_a , the relevant molar mass.
 (b) Estimate the change in temperature when we ascend a height of one kilometer ?
 (c) Show that pressure will depend on height as given by Eq. (1). Find an explicit expression for exponent α in terms of γ .
 (d) According to this model what is the height to which the atmosphere extends? Take $T_0 = 300$ K and $p_0 = 1$ atm.



24. It is well known that the temperature of a closed room goes up if the refrigerator is switched on inside it. A refrigerator compartment set to temperature T_c is turned on inside a hut in Leh (Ladakh). The atmosphere (outside the hut) can be considered to be a vast reservoir at constant temperature T_o . Walls of hut and refrigerator compartment are conducting. The temperature of the refrigerator compartment is maintained at T_c with the help of a compressor engine. We explain the working of the refrigerator engine and the heat flow with the help of the associated figure.



The larger square is the refrigerator compartment with heat leak per unit time Q_c into it from the room. The same heat per unit time Q_c is pumped out of it by the engine (also called compressor and indicated by the smaller square in thick). The compressor does work W and rejects heat per unit time Q_h into the hut. The thermal conductance (in units of watt per kelvin) of the walls of the compartment and hut respectively are K_c and K_h . After a long time it is found that temperature of the hut is T_H . The compressor works as a reverse Carnot engine and it does not participate in heat conduction process.

[Olympiad 2014]

- State the law of heat conduction for the walls of the hut and the refrigerator compartment.
- We define the dimensionless quantities $k = K_h/K_c$, $h = T_H/T_o$ and $c = T_c/T_o$. Express h in terms of c and k .
- Calculate stable temperature T_H given $T_o = 280.0$ K, $T_c = 252.0$ K and $k = 0.90$.
- Now another identical refrigerator is put inside the hut. T_c and T_o do not change but T_H , the hut temperature will change to T'_H . State laws of heat conduction for hut and one of the two identical refrigerator compartments.
- Assume that dimensionless quantities k and c do not change. Let $h' = T'_H/T_o$. Obtain an expression for h' .

HLP Answers

- $n = \frac{\ln \eta}{\ln(1 + \Delta V/V)}$
- $p = p_0 e^{-Ct/V}$
- (a) $T_{\max} = \frac{2}{3}(p_0/R)\sqrt{p_0/3\alpha}$ (b) $T_{\max} = p_0/\epsilon\beta R$
- $\frac{2q}{5(mg + P_0A)}$
- $W = p_0V_0 \ln[(\eta + 1)^2/4\eta]$
- (i) $300\left(\frac{1}{2}\right)^{2/3}$ K (ii) $7500(2^{-2/3} - 1)$ J (iii) $-7500(2^{-2/3} - 1)$ J
- 55 J
- (i) $T_A = 120$ K, $T_B = 241$ K, $T_C = 481$ K, $T_D = 241$ K, (ii) No, (iii) $\Delta Q_{ABC} = \frac{13}{4} \times 10^6$ J ; $\Delta Q_{ADC} = \frac{11}{4} \times 10^6$ J
- $K = 2000$ N/m, $Q = 923$ Joules approx.
- $P = \frac{10}{3} \times 10^5$ N/m², $T = \frac{10500}{31}$ K ≈ 338.71 K
- $T_B = 600 \times 2^{3/5}$ K, $T_D = 1200 \times 2^{-3/5}$ K, $\eta = 61.37\%$
- (a) P_0V_0 (b) $5/2P_0V_0$, $3P_0V_0$ (c) $1/2P_0V_0$ (d) $25P_0V_0/8R$
- $$W = \frac{3}{2} P_1 V_1 \left[1 - \left(\frac{V_1}{V_2} \right)^{2/3} \right]; \Delta U = Q - \frac{3}{2} P_1 V_1 \left[1 - \left(\frac{V_1}{V_2} \right)^{2/3} \right]; T_{\text{final}} = \frac{Q}{3R} + \frac{P_1 V_1}{2R} \left(\frac{V_1}{V_2} \right)^{2/3}$$

$$= \left[\frac{P_1 V_1^{5/3} \cdot V_2^{-2/3}}{2R} + \frac{Q}{3R} \right]$$
- 400 J, 24 K
- (a) $2T_o$, $\frac{p_o}{2}$ in the vessel A and $\frac{T_o}{2^{\gamma-2}}$, $p_o/2^\gamma$ in vessel B, (b) $2T_o$, $P_o/2$
- (a) $\frac{2p_1^{1/\gamma} V_o}{A}$, $\frac{2p_2^{1/\gamma} V_o}{A}$, (b) zero, (c) $(A/2)^\gamma$ where $A = p_1^{1/\gamma} + p_2^{1/\gamma}$
- $\ell = \frac{21840}{673}$ cm



18. (a) zero (b) $\frac{3P_0}{2}$ on the left and $\frac{3P_0}{2}$ on the right (c) $\frac{3T_0}{2}$ (d) $\frac{3P_0V_0}{4}$

19. $\frac{V_0}{2^{\frac{\gamma+1}{\gamma}}}$ in each cases

20. Work done by gas A in isothermal process $W_A = P_0 V_0 \ln \left(\frac{4V_0}{V_0} \right) = 2P_0 V_0 \ln 2$

$$\text{Work done by gas B in adiabatic process } W_B = \frac{P_i V_i - P_f V_f}{\gamma - 1} = \frac{P_0 V_0 - P_0 (2)^{-\gamma} 2V_0}{\gamma - 1} = \frac{P_0 V_0 (1 - 2^{1-\gamma})}{\gamma - 1}$$

According to question. $W_A = 2W_B$

$$2P_0 V_0 \ln 2 = \frac{2P_0 V_0 (1 - 2^{1-\gamma})}{\gamma - 1} \Rightarrow 1 - 2^{1-\gamma} = (\gamma - 1) \ln 2$$

21. (a) $P_1 = \frac{R\alpha T_0}{V_0}$, $P_2 = \frac{\alpha^{\frac{\gamma}{\gamma-1}} R T_0}{n V_0}$, $P_3 = \frac{R T_0}{n V_0}$, $P_4 = \frac{R T_0}{\alpha^{\frac{1}{\gamma-1}} V_0}$
- $V_1 = V_0$, $V_2 = \frac{n V_0}{\alpha^{\frac{1}{\gamma-1}}}$, $V_3 = n V_0$, $V_4 = \alpha^{\frac{1}{\gamma-1}} V_0$
- $T_1 = \alpha T_0$, $T_2 = \alpha T_0$, $T_3 = T_0$, $T_4 = T_0$
- (b) $W_{12} = R \alpha T_0 \ln \left(\frac{V_2}{V_1} \right) = \alpha R T_0 \ln \left(\frac{n}{\alpha^{\frac{1}{\gamma-1}}} \right)$; $W_{23} = -\frac{R}{\gamma-1} (T_0 - \alpha T_0)$
- $W_{34} = R T_0 \ln \left(\frac{\alpha^{\frac{1}{\gamma-1}}}{n} \right)$; $W_{41} = -\frac{R}{\gamma-1} (\alpha T_0 - T_0)$; (c) $Q = R T_0 (\alpha - 1) \ln \left(\frac{n}{\alpha^{\frac{1}{\gamma-1}}} \right)$.
22. 12
23. (a) $\Gamma = \frac{dT}{dz} = \left(\frac{\rho T}{P} \right) g \frac{\gamma-1}{\gamma} = \frac{m_a}{R} g \frac{(\gamma-1)}{\gamma}$ (b) Change in temperature = 9.9 Kelvin
- (c) Comparing $\alpha = \frac{\gamma}{\gamma-1}$ (d) $z = \frac{T_0}{\Gamma} = \frac{300}{9.9} = 30.3 \text{ km}$
24. (a) For hut : $Q_H - Q_C = K_H (T_H - T_0)$. For refrigerator compartment : $Q_C = K_C (T_H - T_C)$
- (b) $h^2 - h(2c + kc) + c^2 + kc = 0$; $h = \frac{(2c + kc) \pm \sqrt{(2c + kc)^2 - 4(c^2 + kc)}}{2}$
- (c) $h = 1.02$ (choosing - sign) $\Rightarrow T_H = 284.7K$
- (d) For hut : $2(Q'_H - Q'_H) = K_H (T'_H - T_0)$
For refrigerator compartment : $Q'_C = K_C (T'_H - T_C)$
- (e) $h'^2 - h' \left(2c + \frac{k}{2}c \right) + c^2 + \frac{k}{2}c = 0$
- $h' = \frac{\left(2c + \frac{k}{2}c \right) \pm \sqrt{\left(2c + \frac{k}{2}c \right)^2 - 4 \left(c^2 + \frac{k}{2}c \right)}}{2}$